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Linear Speed Control For DC Motor

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Here is a simple, low-cost project to control the speed of a DC motor linearly for specific and precise applications. Normally, pulse-width modulation (PWM) is used for varying the speed of DC motors. When width of the pulse is fixed, the speed of motor remains the same, and when its width changes, the speed also changes.

This project uses a low-cost Audino Uno board with few external components. The circuit converts the switch-on time to linear variation in speed with forward-reverse control.

Generally, small DC motors are used to screw-unscrew, drill PCB holes, and for precision grinding, etc. In most of these applications, you need low speed in the beginning that increases with time. Low speed at the beginning enables you to correct the positioning of the workpiece and the tool.

For example, a DC motor used for screwing or unscrewing needs a low speed on starting. If the motor starts at full speed in the beginning itself, you will miss the drive-in hole in the head of the screw, leading to improper screwing or damaging the screw's head. Similarly, drilling holes into a PCB may require low speed in the beginning that increases with time.

Generally, a rotary potentiometer is used to control speed of a DC motor as shown in the module in Fig. 1. But with this proposed speed controller the user can concentrate on drilling or screwing rather than on varying the speed with potentiometer.

Also, there are wear and tear issues in a linear potentiometer that lead to improper contact and malfunctioning of the device after some time. Potentiometers are good but have following fundamental problems:

1. Due to rapid cycling, the inner materials wear out due to friction

2. The wiper movement causes 'fader scratch' noise

3. In some cases, linear/sliding potentiometers (high-end drill machine trigger switch) are used for speed control at the trigger point of a motor. In such cases, speed of the motor is proportional to the depth or pressure at the triggering point. An example of speed control tigger switch used in electric hand drilling machine is shown in Fig. 2.

Fig. 1: DC motor speed controller module

Fig. 2: Speed control trigger switch used in electric hand drilling machine

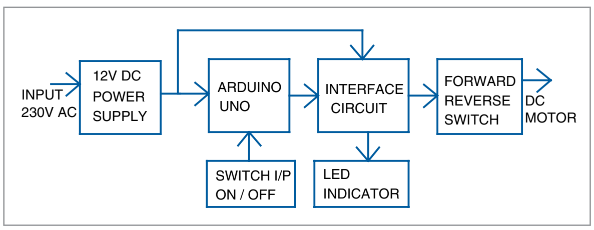

Fig. 3: Block diagram of the time based speed control switch

This circuit converts the switch-on time to speed of the DC motor. So, the speed of the motor increases linearly from zero, till you keep the on/off switch pressed, to its maximum speed.

Circuit and working

Fig. 3 shows block diagram of the time based speed control switch for DC motor. A 12V DC power supply with rated current sufficient to drive the Arduino Uno board and motor is used. The supply is connected through 12mm power jack to Arduino board.

Slow incremental speed of a DC motor is difficult to achieve using discrete components and timing ICs. But here simple on/off pushbutton switch S1 is connected as an input. LED1 connected to pin 9 of Arduino acts as speed indicator. As you continue pressing the switch, the brightness of LED1 increases and so does the speed of DC motor.

The circuit interfaces the PWM signal generated from the Arduino Uno to the 12V DC motor that is being driven. Forward-reverse switch S2 is used for changing the direction of rotation of the motor.

Fig. 4 shows circuit diagram of the time based speed control switch for DC motor. Pin 2 of Arduino Uno Board1 is connected to 5V/IOREF pin through pushbutton switch S1. Digital pin 2 of Arduino and switch S1 are connected to ground through common 10-kilo-ohm resistor R3. Pin 9 of Arduino Uno acts as the output pin, which is also connected to gate of IRF540 MOSFET (T1). MOSFET suits best in PWM applications rather than a transistor.